CONTROL ENGINEERING

1. a) Let

$$A = \begin{bmatrix} -5 & -9 \\ 4 & 7 \end{bmatrix}$$

The characteristic polynomial of this matrix is

$$p(\lambda) = \lambda^2 - 2\lambda + 1 = (\lambda - 1)^2.$$

Thus, the matrix has two identical eigenvalues at 1. To check whether the system is stable or unstable, we need to check the degree m_1 of the eigenvalue in the minimal polynomial. To this end, we check if $\lambda - 1$ is a zeroing polynomial. Obviously,

$$A - I \neq 0.$$

Thus, the minimal polynomial of A is $p(\lambda)$. The multiplicity m_1 of the eigenvalue 1 in the minimal polynomial is 2. Consequently, the equilibrium point of the system is unstable. [5 marks]

b) Let

$$A = \begin{bmatrix} 0 & 1 \\ -\frac{K}{M} & -\frac{F}{M} \end{bmatrix}, \qquad B = \begin{bmatrix} 0 \\ \frac{1}{M} \end{bmatrix}$$

and note that if $K = 0$, then $\det(A) = 0$. Thus, we need to distinguish two cases. If $K \neq 0$, then the equilibrium points are all the points $x_E = -A^{-1}B\bar{u} = \begin{bmatrix} \frac{\bar{u}}{K} & 0 \end{bmatrix}^\top$. If $K = 0$, we need to check if $B\bar{u} \in \operatorname{Im} A$. Since in this case

$$\operatorname{Im} A = \operatorname{Im}\left(\begin{bmatrix} 1 \\ -\frac{F}{M} \end{bmatrix} \right),$$

we clearly see that

$$\begin{bmatrix} 0 \\ \frac{\bar{u}}{M} \end{bmatrix} \notin \operatorname{Im}\left(\begin{bmatrix} 1 \\ -\frac{F}{M} \end{bmatrix} \right)$$

unless $\bar{u} = 0$. In such case, the equilibrium points are all those that satisfy $0 = Ax_E$, which yields $x_E = [\alpha \;\; 0]^\top$ for any $\alpha \in \mathbb{R}$. [5 marks]

c) We determine the characteristic polynomial, obtaining $p(\lambda) = \lambda^2 + \frac{F}{M}\lambda + \frac{K}{M}$. The roots of this polynomial are

$$\frac{-\frac{F}{M} \pm \sqrt{\frac{F^2 - 4KM}{M^2}}}{2}.$$

i) The roots are $\pm j\sqrt{K/M}$. The system is stable.

ii) The roots are complex numbers with negative real part. The system is asymptotically stable.

iii) The roots are $-\frac{F}{2M}$, i.e. two identical negative real numbers. The system is asymptotically stable.

Answers

iv) The roots are two distinct real numbers. Note that $\frac{F}{M} > \sqrt{\frac{F^2 - 4KM}{M^2}}$, thus the roots are negative. The system is asymptotically stable.

v) The roots are 0 and $-\frac{F}{M}$. The system is stable.

vi) In this case the matrix A is

$$A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

The minimal polynomial of this matrix is λ^2 and, consequently, the equilibrium point is polynomially unstable. [6 marks]

d) i) To compute the equilibrium points we need to solve the system

$$\begin{aligned} 0 &= \alpha x_1^3 - x_2 \\ 0 &= \alpha x_2^3 + x_1. \end{aligned}$$

From the first equation we have $x_2 = \alpha x_1^3$, which we substitute in the second equation, yielding

$$0 = \alpha^4 x_1^9 + x_1 = x_1(\alpha^4 x_1^8 + 1).$$

This has only the solution $x_1 = 0$. From $x_2 = \alpha x_1^3$, this also implies $x_2 = 0$. Thus, the system has only one equilibrium point, namely $(0,0)$. [2 mark]

ii) The linearized system is $\dot{x} = Ax$, with

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

The eigenvalues of A are j and $-j$. This is a critical case and we cannot assess the stability of equilibrium point of the nonlinear system using the linearization. [4 marks]

iii) As hinted, we compute

$$\dot{V} = 2x_1\dot{x}_1 + 2x_2\dot{x}_2 = 2\alpha x_1^4 - 2x_1x_2 + 2\alpha x_2^4 + 2x_1x_2 = 2\alpha(x_1^4 + x_2^4).$$

If $\alpha = 0$, then $x_1^2(t) + x_2^2(t) =$ constant. Thus the trajectories move around in a circle and the zero equilibrium point is stable. If $\alpha < 0$, then V decreases, so the "circle shrinks" and the trajectories converge to zero. Thus, the zero equilibrium point is asymptotically stable. If $\alpha > 0$, then V increases, so the "circle grows" and the trajectories diverge. Thus, the zero equilibrium point is unstable. [4 marks]

e) i) Since $A(k)$ is upper triangular, its eigenvalues are the elements on the diagonal. As a result, the eigenvalues of $A(k)$ are both equal to $\frac{1}{2}$ for all k, hence they are constant and with absolute value smaller than one. [3 marks]

ii) Note that

$$x(k) = A(k-1)A(k-2)\cdots A(1)A(0)x(0),$$

Answers

giving $\Phi(k,0) = A(k-1)A(k-2)\cdots A(1)A(0)$. Direct computation yields

$$\Phi(k,0) = \begin{bmatrix} \left(\frac{1}{2}\right)^k & \sum_{j=0}^{k-1}\left(\frac{1}{2}\right)^j \\ 0 & \left(\frac{1}{2}\right)^k \end{bmatrix},$$

which also satisfies the condition $\Phi(0,0) = I$. [6 marks]

iii) By inspection, it is clear that

$$\lim_{k\to\infty} x(k) = \begin{bmatrix} 2x_2(0) \\ 0 \end{bmatrix}.$$

Thus, if $x_2(0) \neq 0$, then the solutions are bounded but do not converge to zero. Thus, the zero equilibrium point is not attractive and it is not asymptotically stable. If $x_2(0) = 0$, then one has

$$\lim_{k\to\infty} x(k) = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

[5 marks]

Answers

2. a) i) The reachability matrix is

$$P = \begin{bmatrix} B & AB \end{bmatrix} = \begin{bmatrix} d & ad \\ 0 & bd \end{bmatrix}.$$

We easily see that full rank is achieved if and only if $b \neq 0$ and $d \neq 0$. If either b or d is zero, the system is not reachable.

[2 marks]

ii) The observability matrix is

$$Q = \begin{bmatrix} C \\ CA \end{bmatrix} = \begin{bmatrix} 0 & e \\ be & ce \end{bmatrix}.$$

We easily see that full rank is achieved if and only if $b \neq 0$ and $e \neq 0$. If either b or e is zero, the system is not observable.

[2 marks]

b) The system is controllable if and only if

$$\operatorname{rank}\left(\begin{bmatrix} P & A^2 \end{bmatrix}\right) = \operatorname{rank}(P),$$

which yields

$$\operatorname{rank}\left(\begin{bmatrix} d & ad & a^2 & 0 \\ 0 & bd & ab+cb & c^2 \end{bmatrix}\right) = \operatorname{rank}\left(\begin{bmatrix} d & ad \\ 0 & bd \end{bmatrix}\right).$$

We distinguish three cases.

i) If $b \neq 0$ and $d \neq 0$, the system is reachable. This implies that it is also controllable.

ii) If $b = 0$ and $d \neq 0$, $\operatorname{rank}(P) = 1$. If $c = 0$, then $\operatorname{rank}\left(\begin{bmatrix} P & A^2 \end{bmatrix}\right) = 1$, which implies that the system is controllable. If $c \neq 0$, then $\operatorname{rank}\left(\begin{bmatrix} P & A^2 \end{bmatrix}\right) = 2$, which implies that the system is not controllable.

iii) If $d = 0$, $\operatorname{rank}(P) = 0$. If $a = c = 0$, then $\operatorname{rank}\left(\begin{bmatrix} P & A^2 \end{bmatrix}\right) = 0$, which implies that the system is controllable. If $a \neq 0$ or $c \neq 0$, then $\operatorname{rank}\left(\begin{bmatrix} P & A^2 \end{bmatrix}\right)$ is either 1 or 2, which implies that the system is not controllable.

[8 marks]

c) i) Since $b = d = 1$, the system is reachable. Let $x(0) = \begin{bmatrix} x_1(0) & x_2(0) \end{bmatrix}^\top$. Then

$$\begin{bmatrix} x_1(1) \\ x_2(1) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(0) = \begin{bmatrix} u(0) \\ x_1(0)+x_2(0) \end{bmatrix}$$

and

$$\begin{bmatrix} x_1(2) \\ x_2(2) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1(1) \\ x_2(1) \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(1) = \begin{bmatrix} u(1) \\ u(0)+x_1(0)+x_2(0) \end{bmatrix}.$$

Hence, the problem is solved selecting $u(0) = 1 - x_1(0) - x_2(0)$ and $u(1) = 1$.

[2 marks]

ii) The equations

$$y(0) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix}, \qquad y(1) = \begin{bmatrix} 0 & 1 \end{bmatrix} \begin{bmatrix} x_1(1) \\ x_2(1) \end{bmatrix}$$

Answers

imply $x_2(0) = 1$ and $x_2(1) = 2$. Substituting these values in the equation

$$\begin{bmatrix} x_1(1) \\ x_2(1) \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1(0) \\ x_2(0) \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \end{bmatrix} u(0)$$

yields $x_1(1) = 2$ and $x_1(0) = 1$. Finally, $x(2)$ can be computed directly from the state equation, yielding $x_1(2) = 1$ and $x_2(2) = 4$.

[2 marks]

d) Recall from part b) that since $b = 0$, the controllability of the system depends on the value of c.

i) If $c = 0$, then the system is controllable. In fact,

$$\begin{bmatrix} x_1(1) \\ x_2(1) \end{bmatrix} = \begin{bmatrix} x_{10} + u(0) \\ 0 \end{bmatrix}, \qquad \begin{bmatrix} x_1(2) \\ x_2(2) \end{bmatrix} = \begin{bmatrix} x_{10} + u(0) + u(1) \\ 0 \end{bmatrix}.$$

The system can be controlled to zero in one step selecting $u(0) = -x_{10}$ and $u(k) = 0$ for all $k > 0$.

[2 marks]

ii) If $c \neq 0$, then the system is not controllable. However, note that there is still a set of initial states that can be controlled to zero. In fact,

$$\begin{bmatrix} x_1(1) \\ x_2(1) \end{bmatrix} = \begin{bmatrix} x_{10} + u(0) \\ x_{20} \end{bmatrix} \qquad \begin{bmatrix} x_1(2) \\ x_2(2) \end{bmatrix} = \begin{bmatrix} x_{10} + u(0) + u(1) \\ x_{20} \end{bmatrix}$$

implies that any initial state such that $x_{20} = 0$ can be controlled to zero in one step selecting $u(0) = -x_{10}$ and $u(k) = 0$ for all $k > 0$.

[2 marks]

e) i) Let $K = [k_1 \; k_2]$ and note that

$$A + BK = \begin{bmatrix} 1 + k_1 & k_2 \\ 1 & 1 \end{bmatrix}$$

and $\det(\lambda I - (A + BK)) = \lambda^2 - (2 + k_1)\lambda + k_1 - k_2 + 1$. Since we want to place all eigenvalues at $\frac{1}{2}$, we want that the characteristic polynomial is $\lambda^2 - \lambda + \frac{1}{4}$, which yields the equations

$$2 + k_1 = 1, \qquad k_1 - k_2 + 1 = \frac{1}{4}.$$

These two equations have solution $k_1 = -1$ and $k_2 = -\frac{1}{4}$.

[4 marks]

ii) Selecting $u = \alpha K x$ yields

$$A + \alpha BK = \begin{bmatrix} 1 - \alpha & -\frac{\alpha}{4} \\ 1 & 1 \end{bmatrix}$$

and $\det(\lambda I - (A + \alpha BK)) = \lambda^2 + (\alpha - 2)\lambda + 1 - \frac{3}{4}\alpha$. We now check the suggested conditions, namely

$$\begin{cases} 1 - \frac{3}{4}\alpha < 1, \\ 1 - \frac{3}{4}\alpha > -1 + \alpha - 2, \\ 1 - \frac{3}{4}\alpha > -1 - \alpha + 2. \end{cases}$$

Answers

These conditions hold concurrently for $\alpha \in \left(0, \frac{16}{7}\right)$. Hence the faulty actuator does not affect the stability of the closed-loop system.

[6 marks]

Answers

3. a) The observability matrix is

$$\mathscr{O} = \begin{bmatrix} 1 & 1 \\ 0 & 0 \end{bmatrix},$$

which has rank one. Hence the system is not observable. To compute the unobservable mode, note that $\lambda(A) = \{0, -3\}$ and consider the observability pencil

$$\begin{bmatrix} \lambda+2 & -2 \\ -1 & \lambda+1 \\ 1 & 1 \end{bmatrix}.$$

The pencil has rank two for $\lambda = 0$ and rank one for $\lambda = -3$, which is therefore the unobservable mode.
The reachability matrix is

$$\mathscr{R} = \begin{bmatrix} 1 & -2 \\ 0 & 2 \end{bmatrix},$$

which has rank two, hence the system is controllable. [5 marks]

b) Let $K = [k_1 \;\; k_2]$ and note that

$$A + BK = \begin{bmatrix} -2+k_1 & 1+k_2 \\ 2 & -1 \end{bmatrix}.$$

Setting $k_1 = 0$ and $k_2 = -1$ yields the desired eigenvalues. [5 marks]

c) Let $L = [l_1 \;\; l_2]'$ and note that

$$A + LC = \begin{bmatrix} -2+l_1 & 1+l_1 \\ 2+l_2 & -1+l_2 \end{bmatrix}.$$

Setting $l_1 = 0$ and $l_2 = -2$ yields the desired eigenvalues (note that any selection of the form $l_1 = -l_2 - 2$ will give the desired eigenvalues). Finally, note that the problem is solvable, even if the system is not observable, because one of the desired closed-loop eigenvalue coincides with the unobservable mode.

[5 marks]

d) The dynamic output feedback controller is described by the equations

$$\dot{\xi} = (A + BK + LC)\xi - Ly \qquad u = K\xi,$$

which for the given system yields

$$\dot{\xi}_1 = -2\xi_1 \qquad \dot{\xi}_2 = -3\xi_2 + 2y \qquad u = -\xi_2$$

By the separation principle the eigenvalues of the dynamic closed-loop system are $\{-1, -2, -2, -3\}$. [5 marks]

e) Note that the *slowest* eigenvalue is -1, which is an eigenvalue of the controller. Hence, the designer has to re-design the controller. The response of the newly designed system will be limited by the slowest eigenvalue of the observer design: -2. The designer can therefore design a new controller with both eigenvalues at -2. A matrix K which achieves this goal is $K = [-1 \;\; -3/2]$.

[5 marks]

Answers

f) By computing the Laplace transform of $\dot{x} = Ax + Bu$, we obtain $X(s) = (sI - A)^{-1}x(0) + (sI - A)^{-1}BU(s)$. The input-output transfer function is

$$\frac{Y(s)}{U(s)} = C(sI - A)^{-1}B = \frac{1}{s}.$$

The input-state transfer vector is

$$\frac{X(s)}{U(s)} = (sI - A)^{-1}B = \begin{bmatrix} \frac{s-1}{s(s-3)} \\ \frac{-2}{s(s-3)} \end{bmatrix}.$$

From the input-output transfer function we can infer that when $U(s) = 1$, the output $y(t)$ of the system is bounded. However, from the input-state transfer vector we can see that the unstable mode $s = 3$ is present. Thus when $U(s) = 1$, the state $x(t)$ diverges to infinity for any non-zero initial condition despite the output of the system being bounded. Consequently, the system is exponentially unstable. [5 marks]

Answers